

It is time to regulate carcinogenic tobacco-specific nitrosamines in cigarette tobacco



The Family Smoking Prevention and Tobacco Control Act gives the Food and Drug Administration power to regulate tobacco products. This commentary calls for immediate regulation of the carcinogenic tobacco-specific nitrosamines 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone (NNK) and N'-nitrosonornicotine (NNN) in cigarette tobacco as a logical path to cancer prevention. NNK and NNN, powerful carcinogens in laboratory animals, have been evaluated as "carcinogenic to humans" by the International Agency for Research on Cancer. NNK and NNN are present in the tobacco of virtually all marketed cigarettes; levels in cigarette smoke are directly proportional to the amounts in tobacco. The NNK metabolite NNAL, itself a strong carcinogen, is present in the urine of smokers and non-smokers exposed to secondhand smoke. Some of the highest levels of NNK and NNN are found in U.S. products. It is well established that factors such as choice of tobacco blend, agricultural conditions, and processing methods influence levels of NNK and NNN in cigarette tobacco and cigarette smoke. Therefore, it is time to control these factors and produce cigarettes with 100 ppb or less each of NNK and NNN in tobacco, which would result in an approximate 15-20 fold reduction of these carcinogens in the mainstream smoke of popular cigarettes sold in the United States.